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FUEL EJECTOR FOR A FUEL CELL

Abstract

A fuel ejector is disclosed to provide gaseous fuel to a fuel cell. The fuel ejector includes an ejector body having first and second fluid inlets and a mixing volume. A nozzle is removably engaged to the ejector body in axial alignment with the first fluid inlet. The nozzle is axially adjustable in position with shims to position the nozzle inlet at a desired location within the mixing volume. A sleeve may also be selected and positioned in the first inlet to establish a throat diameter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to, and the benefit of the filing date of, India Provisional Application No. 202441015323 filed Mar. 1, 2024, which is incorporated herein by reference.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates generally to fuel ejectors, and more particularly to a fuel ejector for a fuel cell.

BACKGROUND

[0003] Fuel cells can use hydrogen gas as a fuel source to mix with air to form water and electricity via electrolysis in a fuel cell membrane. The electricity is then used as a main power source to rotate an output shaft in order to, for example, propel a vehicle. Unused hydrogen gas, along with water vapor and nitrogen, from the electrolysis process can be recirculated for re-use in the fuel cell. Such recirculation systems typically employ a venturi device such as a fuel ejector to incorporate the recirculated gases into the main gaseous hydrogen fuel supply. However, such venturi devices must be configured with dimensional parameters that are specific to the application in which the fuel cell is to be used in order to generate the required flow velocities and recirculation ratios. Therefore, there remains a need for the unique apparatuses, systems, and techniques disclosed herein.

DISCLOSURE OF ILLUSTRATIVE EMBODIMENTS

[0004] For the purposes of clearly, concisely and exactly describing illustrative embodiments of the present disclosure, the manner, and process of making and using the same, and to enable the practice, making and use of the same, reference will now be made to certain exemplary embodiments, including those illustrated in the figures, and specific language will be used to describe the same. It shall nevertheless be understood that no limitation of the scope of the invention is thereby created and that the invention includes and protects such alterations, modifications, and further applications of the exemplary embodiments as would occur to one skilled in the art.

SUMMARY

[0005] The present disclosure includes a fuel ejector for providing gaseous fuel to a fuel cell system. The fuel cell system uses hydrogen gas as a fuel source along with air, and includes a fuel cell that uses electrolysis to produce electricity. The electrolysis process results in residuals of nitrogen gas, water vapor, and hydrogen gas. In order to preserve the unused hydrogen gas, these residuals are returned to the fuel ejector for entrainment and recirculation with incoming hydrogen gas fuel and then provided to the fuel cell for further electrolysis and production of electricity.

[0006] In an embodiment, the fuel ejector includes an elongated ejector body extending along a longitudinal axis from a first inlet to an outlet end of the ejector body. The ejector body includes a second inlet downstream of the first inlet, and a longitudinally extending fuel passage that defines a mixing volume and fluidly connects the first inlet and the second inlet upstream of the outlet end. The fuel ejector also includes a nozzle removably engaged to the outlet end of the ejector body. The nozzle includes a nozzle inlet positioned within the mixing volume to receive gaseous fuel mixed in the mixing volume. The nozzle further includes a nozzle outlet for outletting the mixed gaseous fuel to the fuel cell. The nozzle is adjustable relative to the ejector body along the longitudinal axis to position the nozzle inlet at a selected location within the mixing volume.

[0007] In an embodiment, the fuel ejector includes the elongated ejector body extending along the longitudinal axis from the first inlet to the outlet end of the ejector body. The ejector body includes the second inlet downstream of the first inlet, and the longitudinally extending fuel passage that

defines the mixing volume and fluidly connects the first inlet and the second inlet upstream of the outlet end. The fuel ejector includes the nozzle engaged to the outlet end of the ejector body. The nozzle includes the nozzle inlet positioned within the mixing volume to receive gaseous fuel mixed in the mixing volume. The nozzle further includes the nozzle outlet for outletting the mixed gaseous fuel to the fuel cell. The fuel ejector also includes a sleeve removably positioned in the fuel passage upstream of the mixing volume. The sleeve defines a throat diameter of the fuel passage in the ejector body.

[0008] This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter. Further embodiments, forms, objects, features, advantages, aspects, and benefits shall become apparent from the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The description herein makes reference to the accompanying drawings wherein like numerals refer to like parts throughout the several views, and wherein:

[0010] FIG. 1 is a simplified schematic diagram of a fuel cell system.

[0011] FIG. 2 is a perspective of an example fuel ejector for the fuel cell system of FIG. 1.

[0012] FIG. 3 is a longitudinal section view illustrating of the example fuel ejector of FIG. 2.

[0013] FIG. 4 is an exploded section view of the fuel ejector of FIG. 2.

[0014] FIG. 5 is a longitudinal section view illustrating various dimension parameters of the fuel ejector of FIG. 2.

[0015] FIG. 6 is an enlarged detail view of a portion of the fuel ejector of FIG. 5.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0016] With reference to FIG. 1, there is illustrated a simplified fuel cell system **10** including a fuel ejector **20** and a fuel cell **12**. Fuel cell **12** generates power by using air and hydrogen for electrolysis in one or more membranes to generate electricity. Residual hydrogen gas along with nitrogen and water vapor from fuel cell **12** are recycled in a recycle loop **14** connected to fuel ejector **20**. Fuel cell system **10** also includes a hydrogen source **16** connected to fuel ejector **20** to provide pressurized hydrogen gas. Fuel ejector **20** entrains the two incoming gaseous flows and provides the mixture to a fuel supply **18** connecting fuel ejector **20** and fuel cell **12**. Further details of fuel ejector **20** are provided with reference to FIGS. 2-6.

[0017] In an embodiment, fuel ejector **20** provides gaseous fuel to fuel cell **12**. Fuel ejector **20** includes an elongated ejector body **22** extending along a longitudinal axis L from a first inlet **24** to an outlet end **28** of ejector body **22**. Ejector body **22** includes a second inlet **26** downstream of first inlet **24** and a longitudinally extending fuel passage **30** that defines a mixing volume **32** and fluidly connects first inlet **24** and second inlet **26** upstream of outlet end **28**. Fuel ejector **20** also includes a nozzle **60** removably engaged to outlet end **28** of ejector body **22**. Nozzle **60** includes a nozzle inlet **62** positioned within mixing volume **32** to receive gaseous fuel mixed in mixing volume **32**. Nozzle **60** further includes a nozzle outlet **64** for outletting the mixed gaseous fuel to fuel cell **12**. Nozzle **60** is adjustable relative to ejector body **22** along longitudinal axis L to position nozzle inlet **62** at a selected location within mixing volume **32**.

[0018] In an embodiment, fuel ejector **20** includes elongated ejector body **22** extending along longitudinal axis L from first inlet **24** to outlet end **28** of ejector body **22**. Ejector body **22** includes second inlet **26** downstream of first inlet **24** and longitudinally extending fuel passage **30** that defines mixing volume **32** and fluidly connects first inlet **24** and second inlet **26** upstream of outlet end **28**. Fuel ejector **20** includes nozzle **60** engaged to outlet end **28** of ejector body **22**. Nozzle **60** includes nozzle inlet **62** positioned within mixing volume **32** to receive gaseous fuel mixed in

mixing volume **32**. Nozzle **60** further includes nozzle outlet **64** for outletting the mixed gaseous fuel to fuel cell **12**. Fuel ejector **20** also includes a sleeve **90** removably positioned in fuel passage **30** upstream of mixing volume **32**. Sleeve **90** defines a throat diameter D_t of fuel passage **30** in ejector body **22**.

[0019] Referring to FIG. 2, ejector **20** includes ejector body **22** with first inlet **24**, second inlet **26**, and outlet end **28**. First inlet **24** and outlet end **28** are aligned along longitudinal axis L, body **22** is elongated along longitudinal axis L, and second inlet **26** is transverse to longitudinal axis L. In an embodiment, second inlet **26** is perpendicular to longitudinal axis L, but non-orthogonal orientations are also contemplated and not precluded.

[0020] In the illustrated embodiment, ejector body **22** includes a tapered portion **34** connecting block shaped inlet portion **36** and block shaped mixing portion **38**. Inlet portion **36** and mixing portion **38** may include holes or other features to facilitate mounting of fuel ejector **20** in fuel cell system **10**. Inlet portion **36** and mixing portion **38** may also include other shapes and configurations, such as cylindrical shapes, spherical shapes, conical shapes, irregular shapes, etc.

[0021] Referring further to FIGS. 3-4, a longitudinal section view and exploded section view of fuel ejector **20** are provided. Various dimensional parameters of ejector body **22**, nozzle **60**, and sleeve **90** are shown on a fuel ejector template **120** in FIGS. 5-6 for ease of reference.

[0022] Fuel passage **30** has a first inlet diameter D_1 at or near first inlet **24**. Fuel passage **30** also includes a tapered diffuser portion **41** having a diffuser length L_b . Tapered diffuser portion **41** extends from first inlet diameter D_1 and is tapered at an inlet angle A_1 to a throat region **40** of fuel passage **30**. Throat region **40** extends from diffuser portion **41** to mixing volume **32**. Throat region **40** provides a constant throat diameter D_m along a throat length L_m upstream of mixing volume **32**. Fuel passage **30** widens at a throat outlet angle A_2 into mixing volume **32**.

[0023] Sleeve **90** is positionable in throat region **40** and includes an inner diameter D_m that provides a desired throat diameter D_m when sleeve **90** is positioned in fuel passage **30** along throat region **40**. Sleeve **90** includes a sleeve length L_s between opposite sleeve ends **94**, **96**. In an embodiment, sleeve length L_s corresponds to throat length L_m . In an embodiment, sleeve **90** is selected from a plurality of sleeves **90** having various inner diameters in order to customize throat diameter D_m of fuel ejector **20**. In other embodiments, sleeve **90** is omitted because throat region **40** of fuel passage **30** provides the desired throat diameter D_m .

[0024] Second inlet **26** includes a tubular configuration defining second inlet passage **42**. Second inlet passage **42** includes a second inlet diameter D_2 configured to admit the recirculated hydrogen gas, nitrogen, and water vapor from fuel cell **12**. Second inlet passage **42** opens into mixing volume **32** so that the recycled hydrogen, nitrogen and water vapor from fuel cell **12** mixes with hydrogen fuel provided by fuel passage **30** from fuel source **16** connected to first inlet **24**.

[0025] Nozzle **60** includes a nozzle body **66** extending from nozzle inlet **62** to nozzle outlet **64**. Nozzle body **66** defines a nozzle passage **68** extending from nozzle inlet **62** to nozzle outlet **64**. Nozzle body **66** includes a sealing portion **70** with circumferential grooves **72**. Circumferential grooves **72** receive respective ones of seals **74**. In an embodiment, seals **74** are ring-shaped elastomer seals that fit within corresponding ones of the grooves **72**. Seals **74** contact an inner wall surface **46** of ejector body **22** to prevent gaseous fuel flow between an outer surface of nozzle body **66** and ejector body **22**. In the illustrated embodiment, two seals **74** are provided. Other embodiments contemplate a single seal **74**, or three or more seals **74**.

[0026] Nozzle body **66** also includes a threaded portion **76** between sealing portion **70** and an outer radially extending flange **78**. Threaded portion **76** can threadingly engage internal threads along ejector body **22** adjacent outlet end **28**. Threaded portion **76** forms an axially facing lip **80** adjacent to and downstream of sealing portion **70**. Outer flange **78** is located outside of ejector body **22**, and is positioned adjacent to or in axial engagement with outlet end **28** of ejector body **22**.

[0027] Ejector body **22** includes an internal radially extending surface **48** facing outlet end **28**. A shim **82** is positioned between radially extending surface **48** and axially facing lip **80** to adjust or

set a longitudinal position of nozzle inlet **62** in mixing volume **32**. In particular, the axial location of nozzle inlet **62** can be controlled to provide the desired performance characteristic by using one or more shims **82** between nozzle **60** and ejector body **22**. Fine tuning and axial adjustment of nozzle **60** along longitudinal axis **L** is thus possible with shims **82** while leakage is prevented with seals **74**.

[0028] Nozzle **60**, ejector body **22**, sleeve **90**, and/or shims **82** can provide a modular assembly or sub-assembly of fuel ejector **20**. For example, the final components of fuel ejector **20** can be selected from a plurality of nozzles **60**, a plurality of ejector bodies **22**, a plurality of sleeves **90**, and/or a plurality of shims **82** having differing dimensional parameters. The selected nozzle **60**, ejector body **22**, sleeve **90**, and/or shim(s) **82** are assembled to provide fuel ejector **20** with the desired fuel flow and mixing characteristics for a particular fuel cell application. For example, a fuel cell **12** with a first kilowatt output will require a fuel ejector **20** with different configuration and/or dimensional parameters than a fuel ejector **20** for a fuel cell with a second, higher kilowatt output due to differing flow rate and/or recirculation ratio requirements.

[0029] For example, an ejector body **22** can be selected that provides a desired throat length L_m , tapered diffuser length L_b , and/or mixing chamber **32** length. The throat length L_m and/or tapered diffuser length L_b provide differing suction, entrainment, and/or recirculation ratio capabilities. Different mixing chamber lengths can provide different mixing volumes and interfaces with nozzle **60**.

[0030] If the throat diameter D_m of the selected ejector body **22** is too large, a sleeve **90** having the desired throat diameter D_m can be selected and placed into fuel passage **30** along the throat region **40**. The sleeve **90** may or may not have a length **92** that corresponds to the throat length L_m of the selected ejector body **22**.

[0031] In addition, or alternatively, a nozzle **60** and/or shim(s) **82** can be selected that provide the desired nozzle performance characteristics. The length of nozzle **60** from lip **80** to nozzle inlet **62** and/or shim(s) **82** can be used to control the spacing **S1** of nozzle inlet **62** from the downstream end **33** of mixing chamber **32**.

[0032] In addition, nozzle **60** includes an orifice **84** adjacent to nozzle inlet **62**. Nozzle inlet **62** defines an inlet opening **86** into orifice **84** that is tapered in a downstream direction at angle A_3 to orifice **84**. Orifice **84** has a constant orifice diameter D_o along orifice length L_t . Nozzle **60** then expands at angle A_4 along an expansion length L_n to an outlet region **88** of nozzle passage **68**. Outlet region **88** has a constant diameter D_3 up to outer flange **78**.

[0033] Nozzle **60** can be selected from a plurality of nozzles **60** having different dimensions for these parameters. Different sizes/configurations for orifice **84** provide different flow velocities, which effects suction and entrainment capabilities. The selected nozzle **60** can then be engaged to the selected ejector body **22**. A sleeve **90** and/or one or more shims **82** may also be employed to provide the final desired dimensional parameters to the assembled fuel ejector **20**.

[0034] The modularity and interchangeability of the various components of fuel ejector **20** also allows for different material to be readily used for different components of fuel ejector **20**, such as the ejector body **22**, nozzle **60**, and/or sleeve **90**. Example materials which can be used for any one or more, or all, of the components in an assembled fuel ejector **20** include, for example, stainless steel, aluminum, and/or plastic. Other materials are also contemplated and not precluded.

[0035] Various aspects of the present disclosure are contemplated. According to one aspect, a fuel ejector for providing gaseous fuel to a fuel cell is provided. The fuel ejector includes an elongated ejector body extending along a longitudinal axis from a first inlet to an outlet end of the ejector body. The ejector body includes a second inlet downstream of the first inlet and a longitudinally extending fuel passage that defines a mixing volume fluidly connecting the first inlet and the second inlet upstream of the outlet end. The fuel ejector also includes a nozzle removably engaged to the outlet end of the ejector body. The nozzle includes a nozzle inlet positioned within the mixing volume to receive gaseous fuel mixed in the mixing volume. The nozzle further includes a

nozzle outlet for outletting the mixed gaseous fuel to the fuel cell. The nozzle is adjustable relative to the ejector body along the longitudinal axis to position the nozzle inlet at a selected location within the mixing volume.

[0036] In an embodiment, a sleeve is removably positioned in the fuel passage upstream of the mixing volume. The sleeve defines a throat diameter of the fuel passage in the ejector body. In a further embodiment, the sleeve is selected from a plurality of sleeves to provide the throat diameter based on a desired power output of the fuel cell.

[0037] In a further embodiment, the ejector body defines a throat length along a portion the fuel passage upstream of the mixing volume, and the sleeve defines a sleeve length that corresponds to the throat length. In yet a further embodiment, the fuel passage is tapered toward the throat diameter upstream of the portion of the fuel passage that defines the throat length.

[0038] In an embodiment, at least one shim is positioned between the ejector body and the nozzle to adjust the position of the nozzle inlet along the longitudinal axis.

[0039] In a further embodiment, the ejector body defines a radially extending surface facing the outlet end of the ejector body, the nozzle includes a radially extending lip, and the shim is positioned between the lip and the radially extending surface. In yet a further embodiment, the nozzle includes an outer radially extending flange portion adjacent to or in axial engagement with the outlet end of the ejector body.

[0040] In an embodiment, at least one seal is positioned between the nozzle and the ejector body. The at least one seal prevents gaseous fuel flow between the nozzle and the ejector body. In a further embodiment, the at least one seal includes at least two seals.

[0041] In an embodiment, the nozzle is threadingly engaged to the outlet end of the ejector body. In an embodiment, the mixing volume extends from a first location upstream of the second inlet to a second location downstream of the nozzle inlet.

[0042] According to another aspect of the present disclosure, a fuel ejector for providing gaseous fuel to a fuel cell is provided. The fuel ejector includes an elongated ejector body extending along a longitudinal axis from a first inlet to an outlet end of the ejector body. The ejector body includes a second inlet downstream of the first inlet and a longitudinally extending fuel passage that defines a mixing volume fluidly connecting the first inlet and the second inlet upstream of the outlet end. The fuel ejector includes a nozzle engaged to the outlet end of the ejector body. The nozzle includes a nozzle inlet positioned within the mixing volume to receive gaseous fuel mixed in the mixing volume. The nozzle further includes a nozzle outlet for outletting the mixed gaseous fuel to the fuel cell. The fuel ejector includes a sleeve removably positioned in the fuel passage upstream of the mixing volume. The sleeve defines a throat diameter of the fuel passage in the ejector body.

[0043] In an embodiment, the nozzle is adjustable relative to the ejector body along the longitudinal axis to position the nozzle inlet at a desired location within the mixing volume. In a further embodiment, at least one shim is positioned between the ejector body and the nozzle to adjust the position of the nozzle inlet along the longitudinal axis.

[0044] In yet a further embodiment, the at least one shim is selected from a plurality of shims that provides a desired position of the nozzle inlet along the longitudinal axis. In yet a further embodiment, at least one seal is positioned between the nozzle and the ejector body. The at least one seal prevents gaseous fuel flow between the nozzle and the ejector body.

[0045] In an embodiment, the sleeve is selected from a plurality of sleeves to provide the throat diameter that is based on a desired power output of the fuel cell. In an embodiment, the sleeve is downstream of the first inlet and upstream of the second inlet. In an embodiment, the second inlet is perpendicular to the fuel passage.

[0046] While illustrative embodiments of the disclosure have been illustrated and described in detail in the drawings and foregoing description, the same is to be considered as illustrative and not restrictive in character, it being understood that only certain exemplary embodiments have been shown and described and that all changes and modifications that come within the spirit of the

claimed inventions are desired to be protected. It should be understood that while the use of words such as preferable, preferably, preferred or more preferred utilized in the description above indicates that the feature so described may be more desirable, it nonetheless may not be necessary and embodiments lacking the same may be contemplated as within the scope of the invention, the scope being defined by the claims that follow. In reading the claims, it is intended that when words such as “a,” “an,” “at least one,” or “at least one portion” are used there is no intention to limit the claim to only one item unless specifically stated to the contrary in the claim. When the language “at least a portion” and/or “a portion” is used the item can include a portion and/or the entire item unless specifically stated to the contrary.

Claims

1. A fuel ejector for providing gaseous fuel to a fuel cell, the fuel ejector comprising: an elongated ejector body extending along a longitudinal axis from a first inlet to an outlet end of the ejector body, the ejector body including a second inlet downstream of the first inlet and a longitudinally extending fuel passage that defines a mixing volume fluidly connecting the first inlet and the second inlet upstream of the outlet end; and a nozzle removably engaged to the outlet end of the ejector body, the nozzle including a nozzle inlet positioned within the mixing volume to receive gaseous fuel mixed in the mixing volume, the nozzle further including a nozzle outlet for outletting the mixed gaseous fuel to the fuel cell, the nozzle being adjustable relative to the ejector body along the longitudinal axis to position the nozzle inlet at a selected location within the mixing volume.
2. The fuel ejector of claim 1, comprising a sleeve removably positioned in the fuel passage upstream of the mixing volume, wherein the sleeve defines a throat diameter of the fuel passage in the ejector body.
3. The fuel ejector of claim 2, wherein the sleeve is selected from a plurality of sleeves to provide the throat diameter based on a desired power output of the fuel cell.
4. The fuel ejector of claim 2, wherein: the ejector body defines a throat length along a portion the fuel passage upstream of the mixing volume; and the sleeve defines a sleeve length that corresponds to the throat length.
5. The fuel ejector of claim 4, wherein the fuel passage is tapered toward the throat diameter upstream of the portion of the fuel passage that defines the throat length.
6. The fuel ejector of claim 1, comprising at least one shim between the ejector body and the nozzle to adjust the position of the nozzle inlet along the longitudinal axis.
7. The fuel ejector of claim 6, wherein: the ejector body defines a radially extending surface facing the outlet end of the ejector body; the nozzle includes a radially extending lip; and the shim is positioned between the lip and the radially extending surface.
8. The fuel ejector of claim 7, wherein the nozzle includes an outer radially extending flange portion adjacent to or in axial engagement with the outlet end of the ejector body.
9. The fuel ejector of claim 1, comprising at least one seal between the nozzle and the ejector body, the at least one seal preventing gaseous fuel flow between the nozzle and the ejector body.
10. The fuel ejector of claim 9, wherein the at least one seal includes at least two seals.
11. The fuel ejector of claim 1, wherein the nozzle is threadingly engaged to the outlet end of the ejector body.
12. The fuel ejector of claim 1, wherein the mixing volume extends from a first location upstream of the second inlet to a second location downstream of the nozzle inlet.
13. A fuel ejector for providing gaseous fuel to a fuel cell, the fuel ejector comprising: an elongated ejector body extending along a longitudinal axis from a first inlet to an outlet end of the ejector body, the ejector body including a second inlet downstream of the first inlet and a longitudinally extending fuel passage that defines a mixing volume fluidly connecting the first inlet and the

second inlet upstream of the outlet end; a nozzle engaged to the outlet end of the ejector body, the nozzle including a nozzle inlet positioned within the mixing volume to receive gaseous fuel mixed in the mixing volume, the nozzle further including a nozzle outlet for outletting the mixed gaseous fuel to the fuel cell; and a sleeve removably positioned in the fuel passage upstream of the mixing volume, wherein the sleeve defines a throat diameter of the fuel passage in the ejector body.

14. The fuel ejector of claim 13, wherein the nozzle is adjustable relative to the ejector body along the longitudinal axis to position the nozzle inlet at a desired location within the mixing volume.

15. The fuel ejector of claim 14, comprising at least one shim between the ejector body and the nozzle to adjust the position of the nozzle inlet along the longitudinal axis.

16. The fuel ejector of claim 15, wherein the at least one shim is selected from a plurality of shims that provides a desired position of the nozzle inlet along the longitudinal axis.

17. The fuel ejector of claim 15, comprising at least one seal between the nozzle and the ejector body, the at least one seal preventing gaseous fuel flow between the nozzle and the ejector body.

18. The fuel ejector of claim 13, wherein the sleeve is selected from a plurality of sleeves to provide the throat diameter that is based on a desired power output of the fuel cell.

19. The fuel ejector of claim 13, wherein the sleeve is downstream of the first inlet and upstream of the second inlet.

20. The fuel ejector of claim 13, wherein the second inlet is perpendicular to the fuel passage.
